Assignment 11: Random Walks

PHYS 4202 - Computational Physics (Fall 2026)
Due: Tuesday, November 10, 2026 at 10:00 am
Total points: 100

Purpose

This assignment explores **random walks** as a model of diffusion. You will investigate how random microscopic motion produces predictable statistical behavior, how dimensionality changes distributions, and how boundaries introduce bias.

Environment

All work must be done in **Python 3.9 or newer**. You may use the Python standard library, NumPy, and matplotlib. Use `numpy.random.default_rng` and allow a user-specified seed. Create a generator once per script run and pass it to simulation functions. Do not reset the seed inside a trial or step loop. Vectorized random draws and step updates are encouraged but not required. Implement the random walks and boundary rules yourself. Simulate the individual steps. Do not replace a walk with a direct draw of its final position from a known distribution.

Code organization

Your code directory must contain the following Python files:

- `random_walks.py`: reusable random-walk simulation functions, including the boundary rules used in Part 3.
- `dimensionality.py`: runs the simulations, computes statistics, and generates the figures for Parts 1 and 2 using functions imported from `random_walks.py`.
- `boundaries.py`: runs the Part 3 boundary comparisons and generates their figures using functions imported from `random_walks.py`.

Put reusable simulation logic in functions and allow the main parameters to be changed without editing those functions. Additional helper modules are optional. Include any modules used by the required scripts in your repository. Type hints are encouraged but not required.

Validate inputs and raise helpful errors for invalid values. Step counts and trial counts must be positive integers, and transmission probabilities must satisfy $0 \leq p \leq 1$. Put run instructions and other documentation in the report as described below. A separate README is not required.

Overview

You will simulate random walks in one, two, and three dimensions. The goal is to connect **microscopic randomness** to **macroscopic patterns**.

Choose a number of trials N large enough to produce smooth histograms and stable statistics. You should test a few values of N and ensure your results do not change significantly as N increases. In most cases, this will require on the order of 10^3–10^4 trials.

Part 1: 1D random walk (25 points)

Consider a walker on a line starting at $x = 0$. At each step, the walker moves

$$\Delta x = \pm 1$$

with equal probability.

Write a Python program that:

- simulates a walk of M steps,
- repeats the simulation for N independent trials,
- records the final position of each trial.

Use at least five values of M spanning roughly 100 to 10000.

For this part, create:

- a histogram of final positions for each chosen value of M,
- a calculation of the standard deviation σ of the final positions, computed from the full dataset,
- a plot of σ as a function of M.

Your results should show the diffusion-like scaling

$$\sigma \propto \sqrt{M}.$$

Part 2: Higher-dimensional random walks (30 points)

In all cases below, the step size should be fixed to 1 so results can be directly compared with the 1D walk. For the 2D and 3D distribution plots below, use a single fixed walk length $M = 1000$. This includes the 2D position density and all 1D histograms of final coordinate values and distance r. Each histogram shows the distribution across many independent trials after 1000 steps. Here, "1D histogram" describes the plot of one quantity, even when the walk itself is two- or three-dimensional. Vary M only for the trend plots in the dimensional comparison subsection.

2D walk

At each step:

- draw two random numbers from a normal (Gaussian) distribution,
- treat them as the components of a vector,
- normalize that vector so its magnitude is 1.

This produces a random direction with no preferred orientation.

Simulate many independent trials and record the final position (x, y). Compute

$$r = \sqrt{x^2 + y^2}.$$

For $M = 1000$, create:

- a 2D density plot or 2D histogram of final positions,
- a 1D histogram of final x values,
- a 1D histogram of final y values,
- a 1D histogram of r.

3D walk

At each step:

- draw three random numbers from a normal (Gaussian) distribution,
- treat them as the components of a vector,
- normalize that vector so its magnitude is 1.

This produces a random direction with no preferred orientation.

Simulate many independent trials and record (x, y, z). Compute

$$r = \sqrt{x^2 + y^2 + z^2}.$$

For $M = 1000$, create:

- a 1D histogram of final x values,
- a 1D histogram of final y values,
- a 1D histogram of final z values,
- a 1D histogram of r.

1D, 2D, and 3D comparison

For the following two trend plots, vary the number of steps M instead of holding it fixed at 1000. Use at least five values of M spanning roughly 100 to 10000 in each dimension, using the same chosen values across dimensions. Compute the statistics from many independent trials at each value of M. The distribution histograms above are required only for $M = 1000$.
Create:

- a plot of the standard deviation of a coordinate (e.g., x) vs M for 1D, 2D, and 3D,
- a plot of the mean distance $\langle r \rangle$ vs M for 1D, 2D, and 3D.

In your report, compare:

- component distributions,
- distance distributions,
- scaling with M,
- differences across dimensions.

Part 3: Bias and boundaries in 1D (30 points)

Use a fixed value

$$M = 1000.$$

Each boundary location specifies a **separate setup and run**. In Cases A, B1, and C, run separate ensembles with a single boundary at $x = 5$, $x = 20$, or $x = 50$. Do not place all three boundaries in the same simulation. Start every trial at $x = 0$. Include a separate unbiased ensemble with no boundary for comparison. Overlay the final-position histograms from these separate runs in the figure for that case. Case B2 uses only $x = 20$. In Case D, keep the single wall at $x = 5$ and use a separate ensemble for each transmission probability.

For all boundary cases, treat $x \leq b$ and $x > b$ as the two sides of a boundary at $x = b$. **Check the proposed step before updating the position.** Moving from $b - 1$ to b is allowed and does not trigger the boundary rule. A proposed step from b to $b + 1$ is an attempted crossing and triggers the rule for that case. For example, with a boundary at $x = 20$, a step from 19 to 20 is allowed. The rule is triggered when the walker is at 20 and proposes a step to 21. For return crossings in Cases C and D, a proposed step from $b + 1$ to b is an attempted crossing in the opposite direction.

Case A: Reflecting walls

Use one reflecting wall per setup, at $x = 5$, $x = 20$, or $x = 50$. If a proposed step would cross the wall, reverse the step and then update the position. For a wall at $x = 20$, an attempted step from 20 to 21 instead moves the walker from 20 to 19. Create a single overlaid histogram showing the distribution of final positions for:

- unbiased walk,
- $x = 5$ wall,
- $x = 20$ wall,
- $x = 50$ wall.

Case B1: Absorbing boundaries, survivors only

Use one absorbing boundary per setup, at $x = 5$, $x = 20$, or $x = 50$. If a proposed step would cross the boundary, end the trial immediately without taking that step. The trial is not counted or included in the histogram. Replace it with a new trial. Continue until you have N surviving trials that complete all M steps for each boundary location. Create an overlaid histogram as above. These histograms describe final positions conditioned on surviving without crossing the boundary.

Case B2: Absorbing boundary, stopped walkers counted

Use a single boundary at $x = 20$. Start N independent trials. At the first attempted crossing before or on step M, stop the walk without taking the proposed step. Record the walker's current position, $x = 20$, where it was when it tried to cross. Count this trial and do not replace it. Reaching 20 alone does not stop the walk. If the walker never attempts to cross, record its position after $M = 1000$ steps. Thus every starting trial contributes exactly one recorded position, whether it is absorbed or survives.

Create one additional figure comparing the recorded-position histogram for this case with the Case B1 survivor-only histogram at $x = 20$. Use the same bin edges and normalize each histogram to unit total probability. Use bins centered on the integer positions with width 1 so the accumulation at $x = 20$ is visible. Report the fraction of the N starting trials that were absorbed. Explain why counting stopped walkers creates a concentration at the stopping position and why excluding them in Case B1 gives a different distribution.

Case C: Cliff with no return

Use one cliff per setup, at $x = 5$, $x = 20$, or $x = 50$. Accept the first proposed step from b to $b + 1$. The walker then continues moving on the $x > b$ side, but any proposed return step from $b + 1$ to b is reflected by reversing its direction. Create an overlaid histogram as above.

Case D: Tunneling through a wall

Replace the reflecting wall with a partially transmitting wall. Whenever a proposed step would cross the wall, draw a fresh uniform random number $u \in [0, 1)$. If $u < p$, accept the step. Otherwise, reflect it by reversing the step's direction. Apply this rule to crossings in **both directions**. For a wall at $x = b$, treat $x \leq b$ and $x > b$ as its two sides. Reaching $x = b$ from the left does not yet count as crossing. Each proposed crossing gets an independent transmission decision. This is a simplified stochastic model inspired by quantum tunneling, rather than a simulation of a quantum wavefunction.

Use a wall at $x = 5$ and $M = 1000$. Create a single overlaid histogram of final positions for

$$p = 0, \quad 0.1, \quad 0.5, \quad 1.$$

Use the same number of independent trials for each probability. Check that $p = 0$ reproduces the reflecting-wall case at $x = 5$ and that $p = 1$ reproduces the unbiased walk, allowing for statistical fluctuations when comparing independent samples. Discuss how changing p affects the final-position distribution and explain the two limiting cases. A walker that passes through the wall continues its trial. It can later cross back using the same transmission rule.

In your report, discuss:

- symmetry breaking,
- skewness and truncation,
- dependence on boundary location,
- the distinction between survivor-only and stopped-walker distributions,
- the effect of transmission probability and the $p = 0$ and $p = 1$ limits.

Figures, short report, and submission (15 points)

Prepare a concise report in LaTeX and submit the compiled PDF to Canvas. The report is an organized presentation of your work, not an additional lengthy assignment. Include your name, a descriptive title, and your GitHub repository URL. Organize the discussion as a connected analysis rather than a numbered list of answers, supported by the requested figures, numerical results, comparisons, and checks.

Include every required figure in the report. Label axes, identify curves, and provide concise captions. Clearly identify the number of steps, number of trials, dimension, boundary location, and transmission probability where relevant. Use common bin edges and consistent normalization for overlaid histograms. Put all documentation in the report, including brief run instructions, dependencies, and parameters used for the reported results. Include example commands for running `dimensionality.py` and `boundaries.py`.
Include the following results and discussion:

- Explain how the measured standard deviations in Part 1 support the $\sigma \propto \sqrt{M}$ scaling and describe any deviations seen in finite samples.
- Summarize your trial-count convergence check from the Overview. State which values of N you tested, what statistics or distributions you compared, and why your chosen trial count is sufficient.
- Include a compact numerical table for the dimensional comparisons, listing dimension, step count M, trial count N, coordinate standard deviation, and mean distance $\langle r \rangle$ for each required comparison run. Identify the coordinate used. In 1D, the distance is $r = |x|$.
- Include the component, distance, and scaling comparisons from Part 2 and the boundary and transmission discussion from Part 3.

Include a brief testing or implementation note explaining how you verified the walk and boundary rules, including the reflecting and unbiased limits of the tunneling case.

Include an **AI-use statement**. If you used AI, name the tools, describe their contributions, and explain how you checked or revised the work. If you did not use AI, say so.

Commit the Python code and any supporting modules to your GitHub repository. Submit the report PDF to Canvas. Do not submit plot files separately or commit them to the repository. Keep generated figures out of version control using `.gitignore`. The figures are assessed within the report.

Grading (100 points)

Part 1: 1D random walk (25 points)

Category	What I look for	Pts
Simulation	Correct 1D steps and independent trials	10
Histograms	Final-position histograms for the required step counts	5
Statistics	Standard deviation computed from the full datasets	5
Scaling	Plot and interpretation of the $\sqrt{M}$ trend	5
Total		25

Part 2: Higher-dimensional random walks (30 points)

Category	What I look for	Pts
2D walk	Normalized steps and component and distance histograms	10
3D walk	Normalized steps and component and distance histograms	10
Dimensional comparison	Scaling plots, numerical summary, and interpretation	10
Total		30

Part 3: Bias and boundaries (30 points)

Category	What I look for	Pts
Reflecting walls	Correct Case A rules and overlaid histogram	5
Survivors only	Case B1 rules, replacement trials, and overlaid histogram	5
Stopped walkers	Case B2 counting, comparison figure, and absorbed fraction	5
No-return cliffs	Correct Case C rules and overlaid histogram	5
Tunneling wall	Bidirectional transmission, histogram, and limiting-case checks	5
Interpretation	Boundary location, asymmetry, and transmission effects	5
Total		30

Report and documentation (15 points)

Category	What I look for	Pts
Presentation	All required figures, clear labels, captions, and organization	5
Discussion	Connected analysis supported by results	5
Documentation	Run instructions, repo URL, testing notes, and AI statement	5
Total		15